

1. Suppose there is a football tournament consisting of 15 teams. Is it possible to design the matches so that all teams play exactly 5 matches?
2. Let A be the adjacency matrix of a directed graph G . What does an entry in the matrix A^k signify?
3. Let B be the incidence matrix of a directed graph G . What does an entry in the matrix BB^T signify?
4. The reverse of a directed graph G is the directed graph G^R obtained from G by reversing all edges. Give a linear-time algorithm for computing the reverse of a graph in adjacency list format.
5. An undirected graph is a tree if it is connected and acyclic.
 - (a) Show that any tree on at least 2 vertices has a vertex of degree 1.
 - (b) Show that any tree on n vertices has exactly $n - 1$ edges.
 - (c) Show that there is a unique path between any two vertices of a tree.
6. Describe an $\mathcal{O}(|V(G)|)$ algorithm that determines whether or not a given undirected graph G contains a cycle.
7. Design a linear-time algorithm which, given an undirected graph G and an edge $e \in E(G)$, determines whether G has a cycle containing e or not.
8. Given an undirected graph G and two vertices u and v of G , describe an algorithm to determine the number of distinct shortest paths from u to v .
9. An undirected graph G is called bipartite if it contains no odd length cycles.
 - (a) Show that if G is bipartite then there are no edges between vertices in the same level of any BFS forest.
 - (b) Given a connected bipartite graph G , modify BFS to find a partition of the vertex set into A and B such that each edge is between some vertex in A and some vertex in B .
10. Consider the following problem. There are three containers whose sizes are 10 litres, 7 litres, and 4 litres, respectively. Initially, the 7-litre and 4-litre containers are completely filled with water while the 10-litre container is empty. Only one type of operation is allowed: pouring the contents of one container into another, stopping only when the source container is empty or the destination container is full. Model the following tasks as graph-theoretic problems: give a precise definition of the graph involved, state the specific question about this graph that needs to be answered and mention which algorithm can be applied to solve the problem.
 - (a) (1 point) How can one determine if there is a sequence of operations that leaves exactly 2 litres in the 7- or 4-litre container?
 - (b) (2 points) Given an additional input $k \in \mathbb{N}$, how can one determine if there is a sequence of at most k operations that leaves exactly 2 litres in the 7- or 4-litre container?